POWER BI REPORT GUIDE

Olympic Games: who wins, and how efficiently

How the Olympics report was built, what each page shows, how to use the dashboards, and how to maintain the project.



Taiwo Ajayi

Report author

Report: Olympics.pbip (18 pages: 14 story pages, 3 dashboards, 1 drill-through)
Data: Project_06_Olympic_Games.xlsx as stored in the model: 30 nations, 17 Games (1992-2022), 26,824 entries
Tools: Power BI Desktop (PBIP format), DAX queries, powerbi-report-author CLI, Node.js
Version: October 2026

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Appendix A. Project files

> **Who this guide is for.** Olympic committee and sport-funding readers who will read or present the report (sections 1, 2, 7, 8 and 9), and analysts who will maintain it (sections 3 to 6 and 10 to 13).

1 · EXECUTIVE SUMMARY

1,036 medals from 26,824 entries, and nine nations win 61% of them

1,036
medals won (334 gold, 316 silver, 386 bronze)

3.9%
of entries win a medal

61%
of medals won by nine nations

6.2
medals per million people in Jamaica, the best rate

Verdict	Topic	Evidence
CONCENTRATED	Nine nations	Australia 82, Korea 76, China 74, Russia 73, France 72, Germany 72, Japan 71, USA 64, Great Britain 47: 631 medals and 207 of 334 golds
EFFICIENT	Conversion	Every nation enters about 900 times; the leaders turn 6-9% of entries into medals, everyone else 1-3%
SURPRISE	Small nations	Per million people: Jamaica 6.2, New Zealand 4.3, Australia 3.2. Per $bn GDP: Jamaica 1.06, Kenya 0.19
NO EFFECT	Home advantage	Hosts won 2.75 medals at home against 3.15 in an average Games; 11 of 16 did worse at home
BALANCED	Women and men	Women won 492 medals and men 546, from 904 and 896 athletes

Recommended actions

1. **Fund conversion, not entries.** Put money where athletes already reach finals. Target: lift mid-table conversion from 2% to 4%.
2. **Benchmark per head.** Judge smaller nations on medals per million people and per $bn GDP, not raw counts.
3. **Fix the data first.** Restore the source workbook and reload a complete medal list before any forecast.

Find what separates medal-winning nations from the rest

The project was briefed as an Olympic Games historical performance explorer: analyse medal performance, country specialisation, athlete demographics, economic and population factors, and future performance. The report answers five questions:

1. How concentrated are medals among nations?
2. Which nations convert entries into medals best?
3. Do population, wealth and region explain success?
4. Has the medal haul changed over time, and do hosts win more at home?
5. Which sports and athletes win medals?

> **Forecasting was dropped.** The original report had participation what-if pages. The data cannot support a forecast (section 13), so the report states what the data shows and what must be fixed first.

One fact table of entries and five lookup tables

The source workbook `C:\Users\USER\Downloads\Project_06_Olympic_Games.xlsx` is missing. The model still holds the imported data, so every number in this guide was checked with DAX queries against the open model.

Table	Grain	Key columns
FactResult	26,824 athlete entries	AthleteID, GamesID, EventID, SportID, NOC_Code, Medal (Gold, Silver, Bronze or NA), DistinctMedalKey
DimNOC	30 nations	CountryName, Region, IncomeGroup, population (millions), GDP ($bn)
DimGames	17 Games	Year, Season, HostCity, HostCountry; new Games Label, Games Order, Host NOC
DimEvent / DimSport	139 events, 20 sports	EventName, IsTeamEvent; SportName, SportFamily
DimAthlete	1,800 athletes	Sex, BirthYear, HeightCm, WeightKg
MedalTable	Calculated summary	Holds the report measures

Data quality

- Only 315 of 2,363 event-Games have any medal, and 67 award the same medal to more than one nation.
- Every sport appears in both Summer and Winter Games.
- Birth years do not match the Games: ages run from -13 to 57.
- 78 athletes have weight 0 and 80 have no height.

A star around FactResult, with a calculated medal summary

From	To	Notes
FactResult[AthleteID]	DimAthlete[AthleteID]	Many to one
FactResult[EventID]	DimEvent[EventID]	Many to one
DimEvent[SportID]	DimSport[SportID]	Sport reaches the fact through the event
FactResult[SportID]	DimSport[SportID]	Inactive
FactResult[GamesID]	DimGames[GamesID]	Many to one
FactResult[NOC_Code]	DimNOC[NOC_Code]	Many to one
MedalTable	FactResult, DimEvent, DimNOC	Calculated table; one link filters both ways

Report measures live in `MedalTable`; those added for this version are in the display folder **Report v2**. Unused what-if parameter tables were left in place.

The measures the report uses

Measure	Definition	Format
Total Medals	Distinct event-Games-nation-medal keys, excluding "NA"	#,0
Gold / Silver / Bronze Medals	Total Medals for one medal colour	0
Entries	Rows in FactResult	#,0
Medal Conversion %	Total Medals / Entries	0.0%
Medal Share %	Share of all medals across nations	0.0%
Medals per Games	Total Medals / Games Count	0.0
Medals Per Million / Per Billion GDP	Total Medals / population (millions) or GDP ($bn)	0.00 / 0.000
Gold Share %	Gold / Total	0.0%
Medallists, Athletes Competing, Nations	Distinct athletes with a medal, athletes, nations	#,0
Top-3 Sport Share	Medals from the three strongest sports / Total Medals	0.0%
Top Nation	Nation with the most medals in the selection	text
Host Medals at Home / Host Average per Games	Host nation's medals at its own Games, against its average Games	0 / 0.0

```
Total Medals =
CALCULATE ( DISTINCTCOUNT ( FactResult[DistinctMedalKey] ),
            KEEPFILTERS ( FactResult[Medal] <> "NA" ) )

Host Medals at Home =
SUMX ( VALUES ( DimGames[GamesID] ),
       VAR _host = CALCULATE ( MAX ( DimGames[Host NOC] ) )
       RETURN CALCULATE ( [Total Medals], FactResult[NOC_Code] = _host ) )
```

KEEPFILTERS matters: without it, Gold Medals (which filters Medal = "Gold") would be overwritten by the "not NA" filter and return every medal.

6 · CORRECTIONS

Five things fixed from the original report

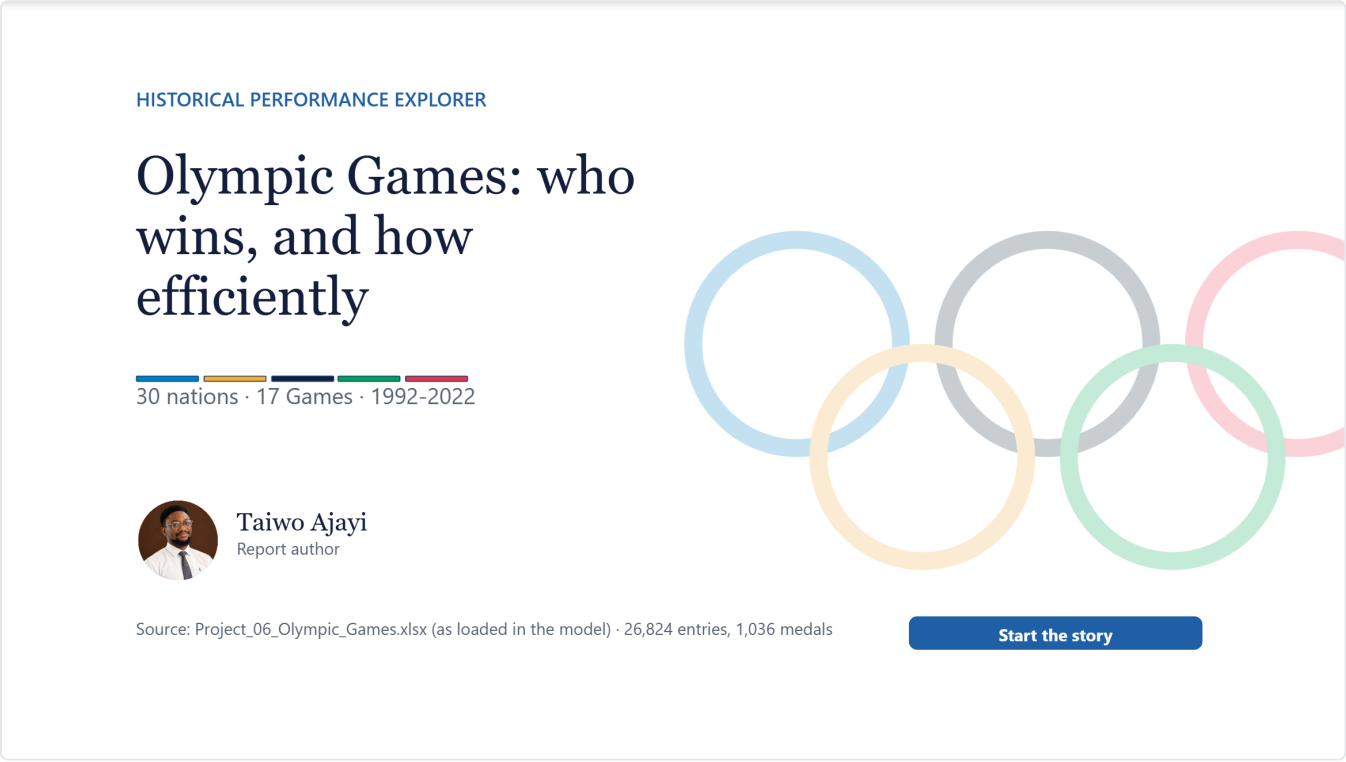
#	Issue	Fix
1	Total Medals counted every entry, including 21,571 "NA" keys, so the report showed 23K medals while gold, silver and bronze added up to 1,036	Medals now exclude "NA": 1,036
2	Medals Per Million divided population (already in millions) by 1,000,000; Medals Per Billion GDP divided GDP (already in $bn) by 1,000,000,000	Extra division removed: Jamaica 6.21 per million, 1.06 per $bn
3	Sport Specialization Index divided a table by a number	Now the share of medals from the top three sports (25.4%)
4	Insight pages said the United States leads	On real medals Australia leads (82); the USA is 8th (64)
5	Forecast pages built on the inflated medal count	Dropped; the data cannot support a forecast

Story pages: fourteen pages, one finding each

Each story page has a Georgia title that states the finding, grey-blue context bars with the key bars highlighted, a source line and a Dashboards button. Gold is used only for gold medals.

1 Cover

Light cover with the Olympic rings drawn large and faded behind the title, a five-colour rule, and the author block. **Start the story** opens page 2.



2 Brief

Aim, five questions, and scope tiles: 30 nations, 17 Games, 1,800 athletes, 26,824 entries.

The brief: find what separates medal-winning nations from the rest

Aim
Help Olympic committees and sport funders see which nations win, how efficiently they turn entries into medals, and which factors go with success.

Questions this report answers

- How concentrated are medals among nations?
- Which nations convert entries into medals best?
- Do population, wealth and region explain success?
- Has the medal haul changed over time, and do hosts win more at home?
- Which sports and athletes win medals?

30
nations

17
Games (8 Summer, 9 Winter)

1,800
athletes

26,824
entries

20 sports and 139 events, Barcelona 1992 to Beijing 2022.
The source workbook is missing; numbers come from the data stored in the model.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 2

Dashboards

3 Scorecard

Four headline KPIs and five findings with verdict pills.

1,036 medals from 26,824 entries, and nine nations win 61% of them

1,036
medals won

334
gold medals

26,824
athlete entries

3.9%
of entries win a medal

CONCENTRATED	Nine nations	Win 631 of the 1,036 medals (61%) and 62% of the golds
EFFICIENT	Conversion	The leaders turn 6-9% of entries into medals; everyone else 1-3%
SURPRISE	Small nations	Jamaica wins 6.2 medals per million people, New Zealand 4.3
NO EFFECT	Home advantage	Hosts won 2.75 medals at home against 3.15 in an average Games
BALANCED	Women and men	Women won 492 medals, men 546, from equal athlete numbers

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 3

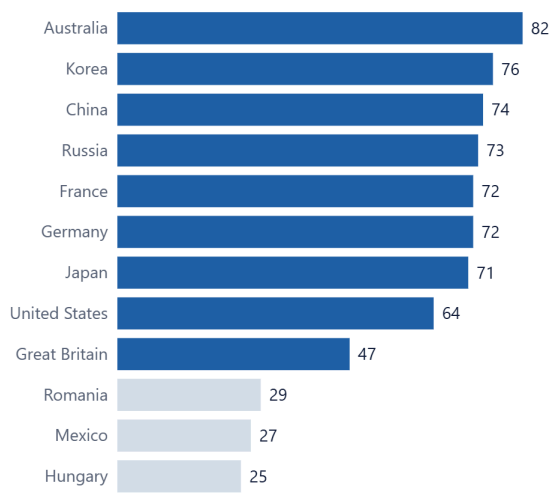
Dashboards

4 Medal table

Medals by nation with the nine leaders highlighted, and a medal table with gold shaded.

Nine nations win 61% of all medals; Australia leads with 82

Medals by nation, top 12



Medal table, top 10

Nation	Gold	Silver	Bronze	Total	Share
Australia	26	31	25	82	7.9%
Korea	20	26	30	76	7.3%
China	23	18	33	74	7.1%
Russia	25	23	25	73	7.0%
France	24	22	26	72	6.9%
Germany	23	20	29	72	6.9%
Japan	27	23	21	71	6.9%
United States	24	20	20	64	6.2%
Great Britain	15	12	20	47	4.5%
Romania	5	10	14	29	2.8%

A clear gap after ninth place.
The nine leaders win 47-82 medals and 207 of the 334 golds; the next nation, Romania, wins 29.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 4

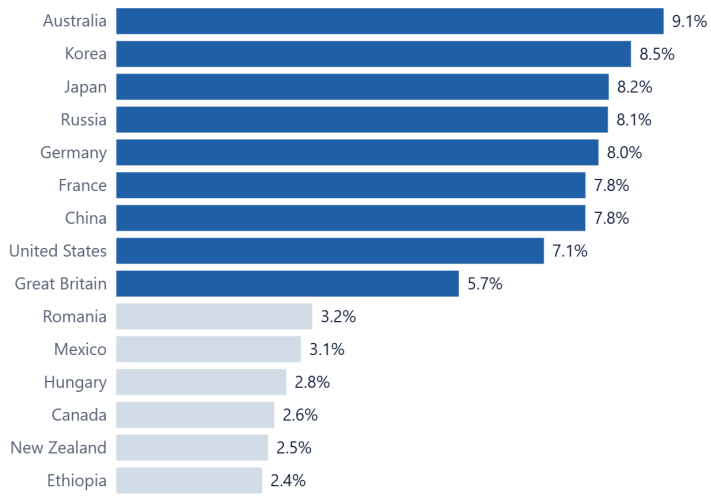
Dashboards

5 Conversion

Medals per 100 entries by nation: 6-9% for the leaders, 1-3% for everyone else.

The leaders turn 6-9% of their entries into medals; every other nation 1-3%

Medals per 100 entries, by nation (top 15)



26,824
entries, about 900 per nation

3.9%
conversion, all nations

Every nation sends about the same number of entries.
The gap is in what those entries achieve, not how many there are.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 5

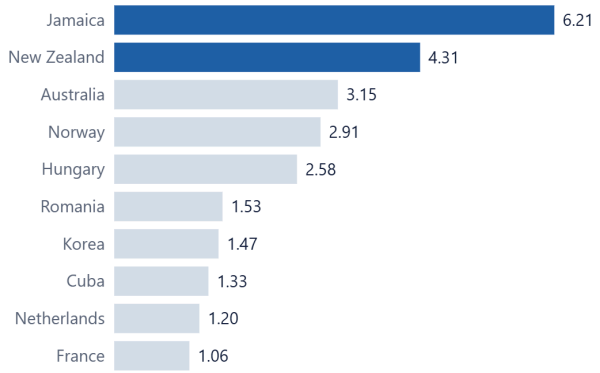
Dashboards

6 Per capita

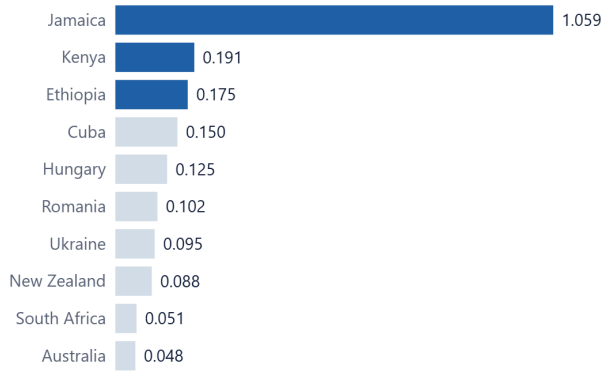
Medals per million people and per $bn of GDP: Jamaica leads both.

Small nations punch above their weight: Jamaica wins 6.2 medals per million people

Medals per million people, top 10



Medals per $bn of GDP, top 10



China and India sit near the bottom on both measures: size alone does not win medals.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · population and GDP at the latest Games · 6

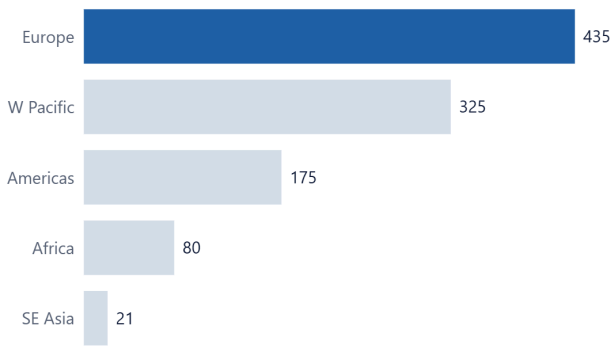
Dashboards

7 Regions

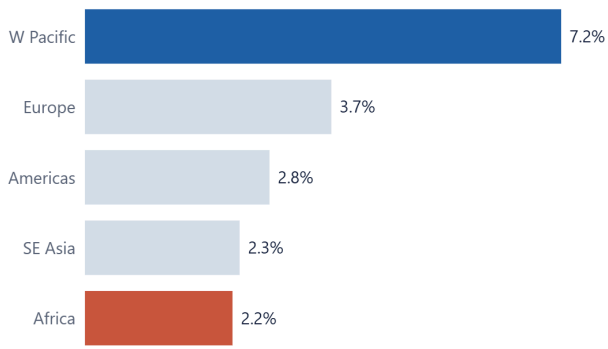
Medals and conversion by region: W Pacific converts best, Africa least.

W Pacific nations convert 7.2% of entries into medals; African nations 2.2%

Medals by region



Medals per 100 entries, by region



Europe wins the most because it sends 13 nations; W Pacific wins more per entry than anyone.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 7

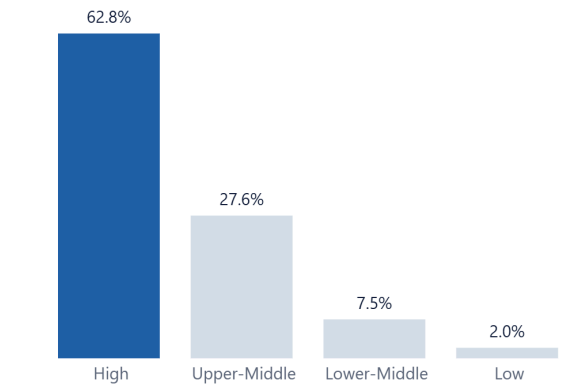
Dashboards

8 Income

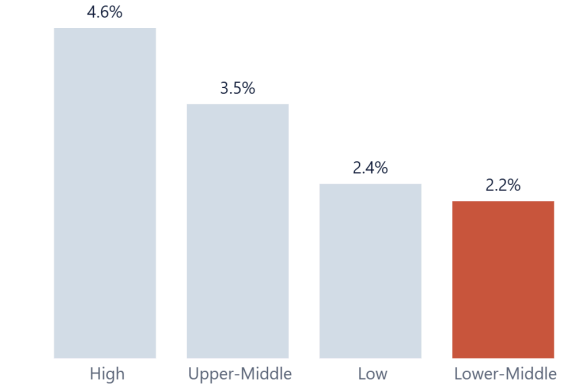
Share of medals and conversion by income group.

High-income nations win 63% of medals and convert twice as well as poorer ones

Share of medals by income group



Medals per 100 entries, by income group



Only one nation (Ethiopia) is low income, so read the Low bar with care.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 8

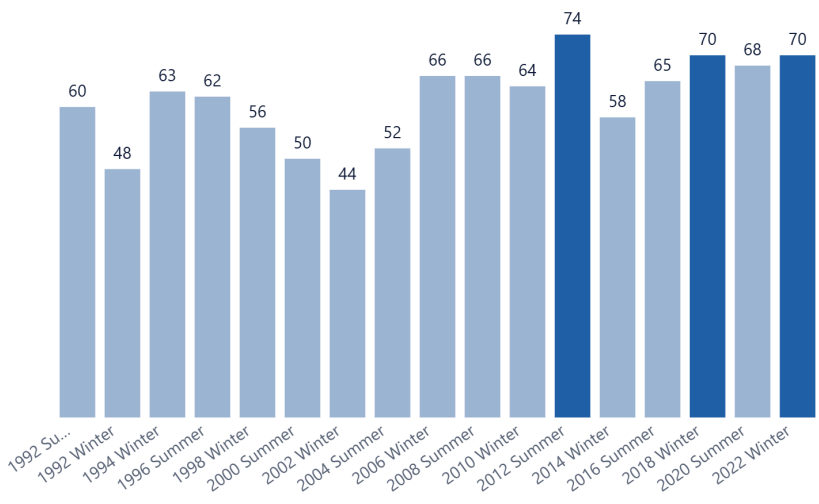
Dashboards

9 Trend

Medals at each Games in order, with the average before and after 2004.

Medals per Games rose from 54 to 67 after 2004

Medals by Games



54.4
medals per Games, 1992-2004

66.8
medals per Games, 2006-2022

Entries rose only 8%.
From 1,512 to 1,636 per Games while medals rose 23%, so each entry became more likely to win.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 9

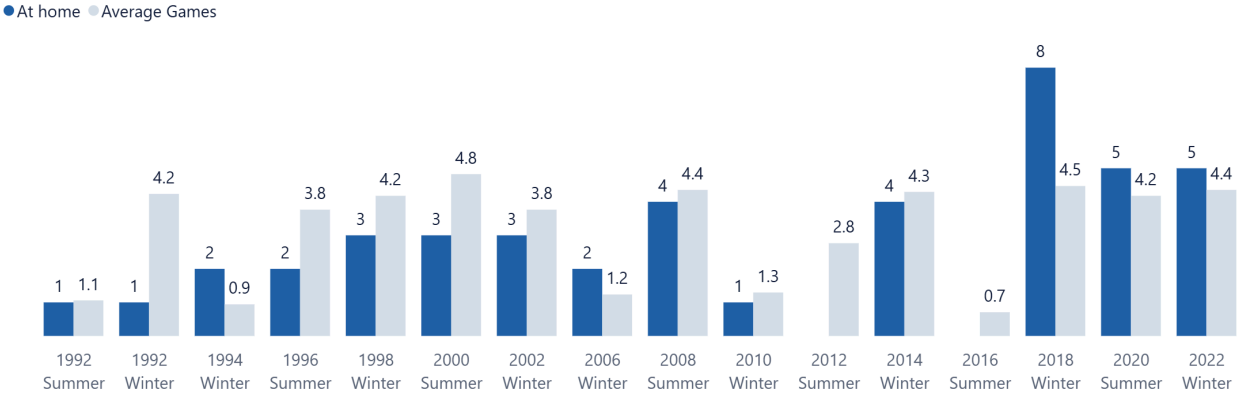
Dashboards

10 Host effect

Each host nation's medals at home against its average Games.

Hosts win no more at home: 2.75 medals per home Games against 3.15 on average

Host nation medals at its home Games vs its average Games



11 of 16 hosts won fewer medals at home than in an average Games; Korea 2018 (8 against 4.5) is the clear exception. Greece (2004) is not in the nation list.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 10

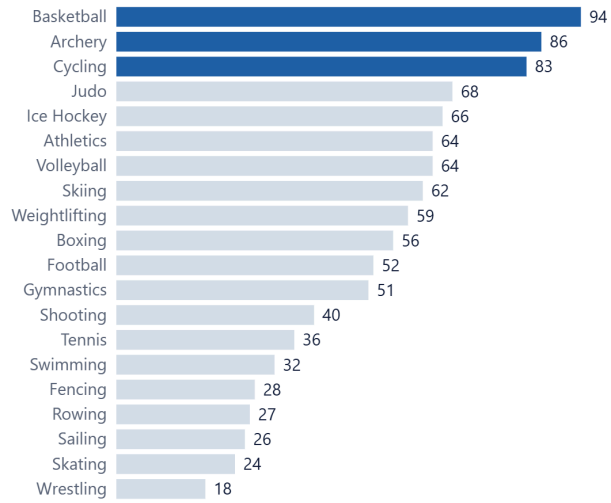
Dashboards

11 Sports

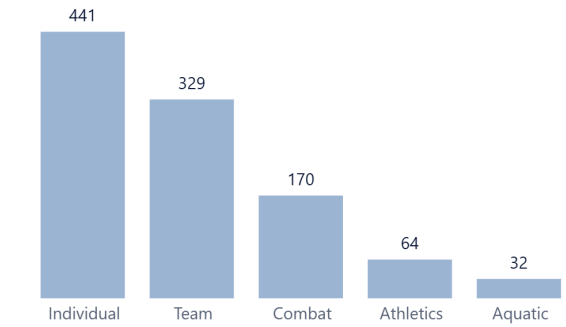
Medals by sport and by sport family, with a note that medals follow the number of events.

Medals follow the number of events: basketball, archery and cycling lead

Medals by sport



Medals by sport family



Basketball holds 12 events, archery 10, cycling 11; wrestling holds 4. Count events before judging a sport as a medal source.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 30 nations, 17 Games, 1992-2022 · 11

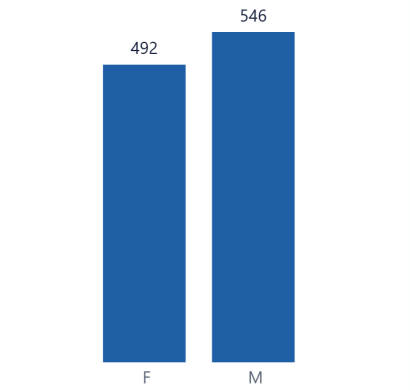
Dashboards

12 Athletes

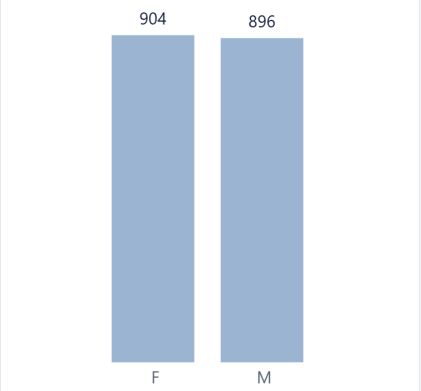
Medals, athletes and conversion by sex, and why age is not analysed.

Women and men win medals almost equally: 492 against 546

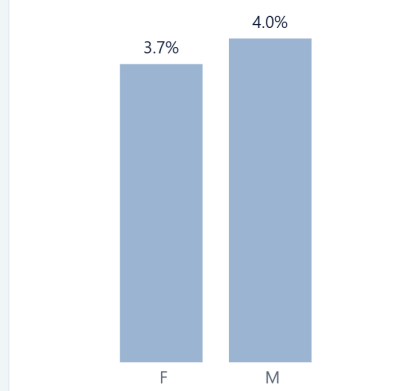
Medals by sex



Athletes by sex



Medals per 100 entries



Age is not analysed: birth years do not match the Games (ages run from -13 to 57), and 78 athletes have weight 0.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · 1,800 athletes · 12

Dashboards

13 Recommendations

Three actions with targets and supporting KPIs.

Three actions: fund conversion, benchmark per head, fix the data

1

Fund conversion, not entries

Every nation enters about 900 times; the leaders win 6-9% of those entries. Put money where athletes already reach finals.

Target: lift mid-table conversion from 2% to 4%

2

Benchmark per head

Judge smaller nations on medals per million people and per $bn GDP, where Jamaica, New Zealand and Kenya lead.

Target: per-capita goals, not raw medal counts

3

Fix the data first

Only 13% of event-Games have medals and birth years do not match the Games. Reload a complete list before forecasting.

Target: medals for all 2,363 event-Games

3.9%
conversion today, all nations

0.21
medals per million people

60.9
medals per Games

Source: Olympics model (Project_06_Olympic_Games.xlsx) · targets are planning goals, not forecasts · 13

Dashboards

14 Data notes

Corrections, data limits, a Summer and Winter comparison, and what still holds.

Assumptions and data issues behind these numbers

Corrections to the original report

- Total Medals counted every entry (23K); it now counts medals only (1,036).
- Medals per million and per $bn GDP divided by a million and a billion twice; fixed.
- The old pages said the USA leads; on true medals Australia leads and the USA is 8th.
- Forecast pages are dropped: the data cannot support a forecast.

Data limits

- Only 315 of 2,363 event-Games have medals, and 67 award the same medal twice.
- Every sport appears in Summer and Winter, so season splits mean little.
- Ages run from -13 to 57 and 78 athletes weigh 0 kg; body and age measures are not used.
- The source workbook is missing, so the data cannot be refreshed.

Summer and Winter side by side

Season	Games	Entries	Medals	Per Games
Summer	8	12,616	497	62.1
Winter	9	14,208	539	59.9

Treat this as a teaching dataset.

The patterns compare nations inside this data. They are not a record of real Olympic history: real Games award a gold, silver and bronze in every event, and sports belong to one season.

What still holds

Comparisons between nations, regions and income groups, because every nation is measured the same way.

What to do next

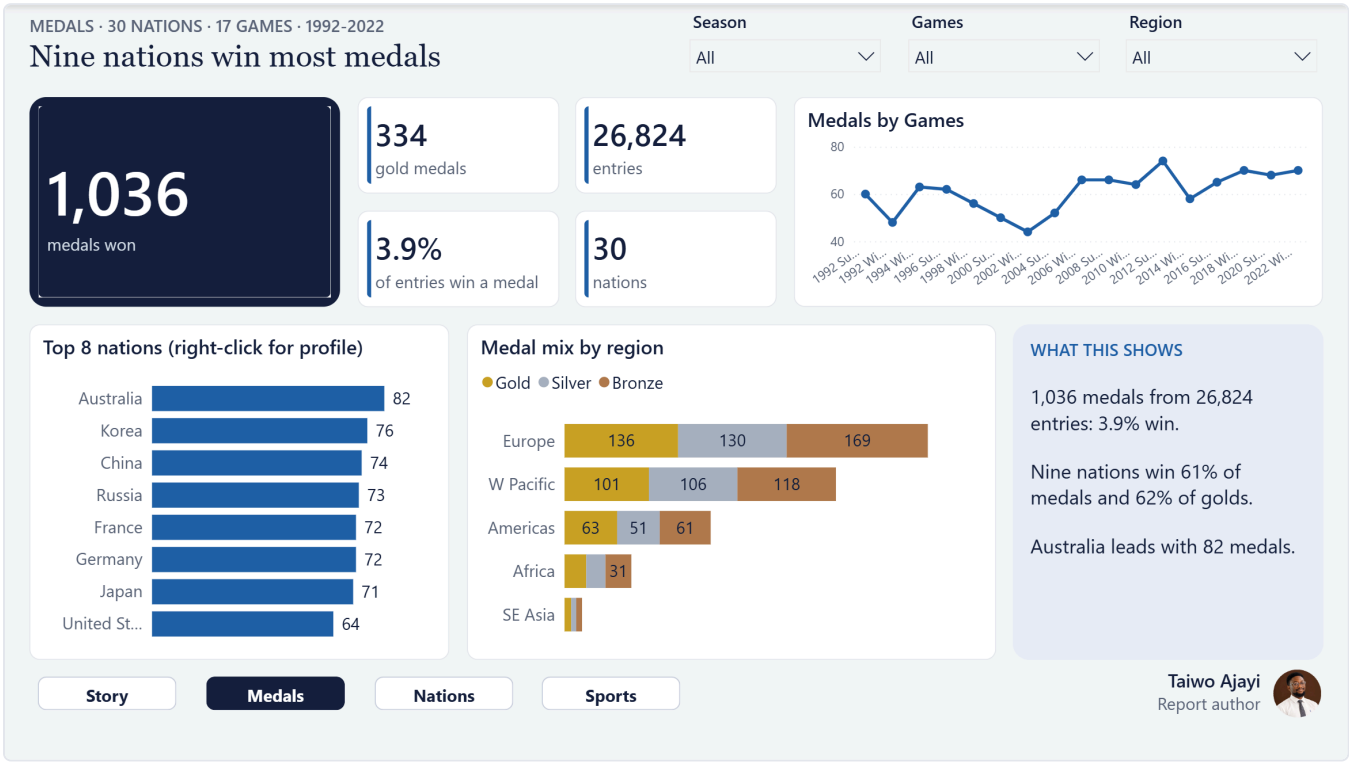
Restore the source workbook and reload a complete medal list before using the report for decisions.

Source: Olympics model (Project_06_Olympic_Games.xlsx) · all entries · 14

Dashboards: three bento pages and a nation profile

D1 Medal overview

Hero tile with total medals. KPI cluster: gold, entries, conversion, nations. Medals by Games, top 8 nations, medal mix by region, and a "what this shows" tile.



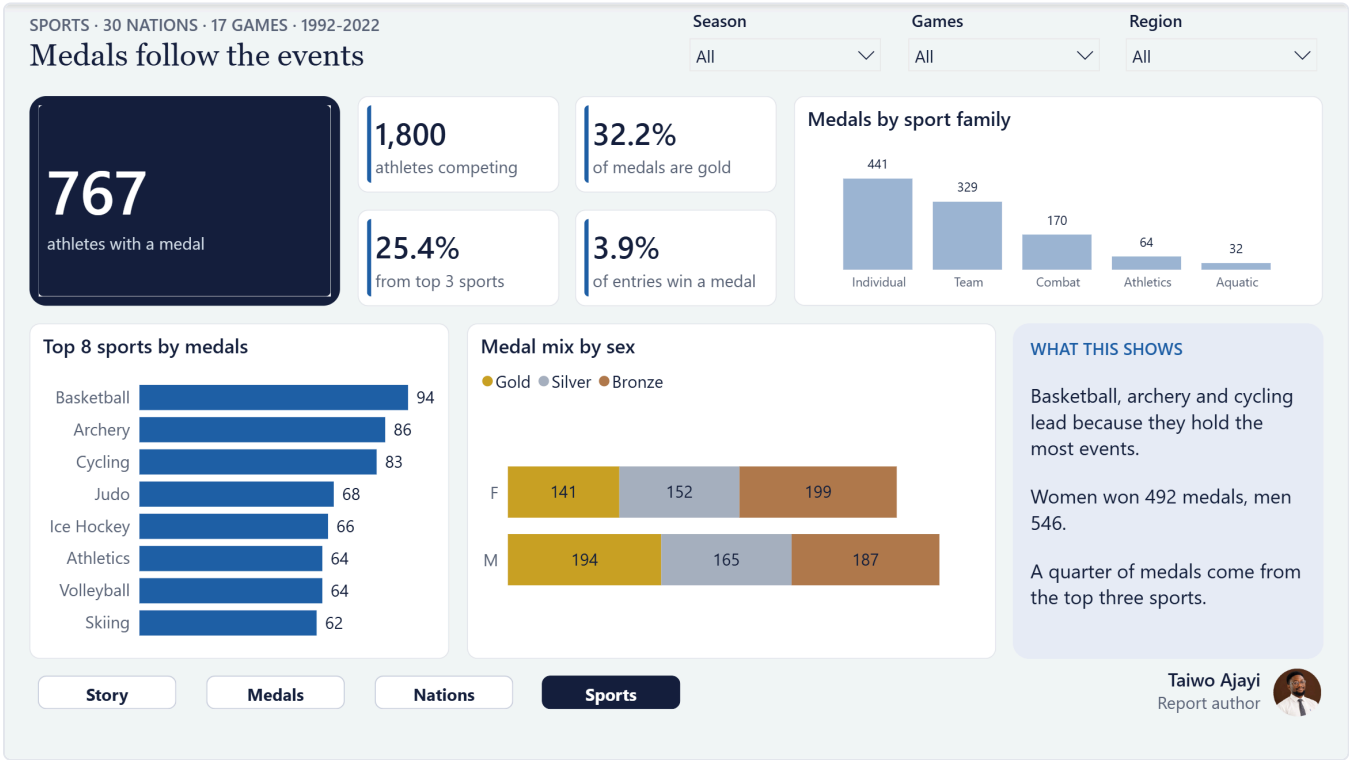
D2 Nations and efficiency

Hero tile names the top nation. KPI cluster: medals per Games, gold share, per million, per $bn GDP. Conversion by region and by nation, and medals per million people.



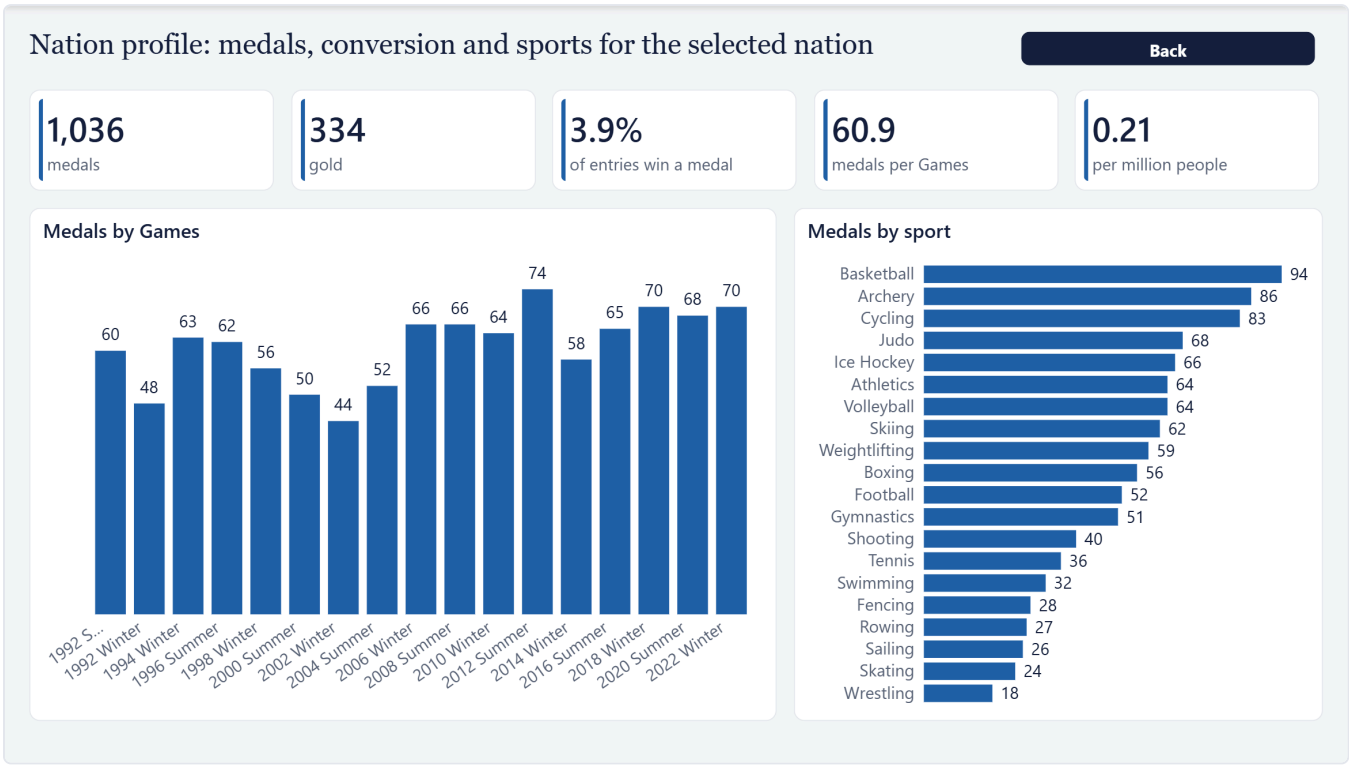
D3 Sports and athletes

Hero tile with medallists. KPI cluster: athletes, gold share, top-3 sport share, conversion. Medals by sport family and by sport, and medal mix by sex.



Nation profile

Hidden drill-through page. Right-click a nation bar and choose Drill through, then Nation profile, for that nation's KPIs, medals by Games and by sport. **Back** returns you to the page you came from.



Reading, filtering, drilling and presenting

Navigating

- Start on the cover and press **Start the story**, or use the page tabs. In Desktop, hold Ctrl when clicking a button.
- Every story page has a **Dashboards** button; the dashboards have a **Story** tab back.

Filtering

- **Season**, **Games** and **Region** filters sit at the top of each dashboard and stay in sync across all three.
- Clicking a bar cross-filters the other visuals on the page; click it again to clear.

Drilling to a nation

- Right-click a nation bar (D1 top nations, D2 conversion) and choose **Drill through → Nation profile**.
- The profile shows that nation's medals, gold, conversion, medals per Games and per million, by Games and by sport. Press **Back** to return.

Presenting

- Use the story pages in order for a briefing; switch to the dashboards for questions.
- Say "in this dataset" when quoting totals: the medal list is incomplete (section 13).

10 · DESIGN SYSTEM

Calm and editorial, with the Olympic rings on the cover

Navy #14213D
titles, hero tiles

Blue #1F5FA8
accent, highlights

Gold #C9A227
gold medals only

Terracotta #C8553D
problems only

- Cover: the five rings drawn large and faded behind the title, with a five-colour rule under it (`_build\make-cover.ps1`).
- Page #F4F5F7; white tiles with a #E3E6EC hairline; Georgia titles; Segoe UI body; nothing under 12pt.
- Medal mixes use gold, silver (#A7B1BF) and bronze (#B07B4F).
- Bento dashboards: dark hero tile, 2 × 2 KPI cluster, wide chart, mixed-width tiles and a tinted "what this shows" tile; author block bottom-right.
- Same design system as the Bank Churn and FAO reports, built from one shared generator.

Tie-out values (no filters)

Check	Value
Medals / gold / silver / bronze	1,036 / 334 / 316 / 386
Entries / conversion	26,824 / 3.9%
Nations / Games / athletes / medallists	30 / 17 / 1,800 / 767
Top nations: Australia / Korea / China / Russia / France / Germany / Japan / USA / GB	82 / 76 / 74 / 73 / 72 / 72 / 71 / 64 / 47
Conversion: Australia / Korea / Japan / Russia	9.1% / 8.5% / 8.2% / 8.1%
Per million: Jamaica / New Zealand / Australia	6.21 / 4.31 / 3.15
Per $bn GDP: Jamaica / Kenya / Ethiopia	1.059 / 0.191 / 0.175
Region medals: Europe / W Pacific / Americas / Africa / SE Asia	435 / 325 / 175 / 80 / 21
Region conversion: W Pacific / Europe / Americas / SE Asia / Africa	7.2% / 3.7% / 2.8% / 2.3% / 2.2%
Income share: High / Upper-Middle / Lower-Middle / Low	62.8% / 27.6% / 7.5% / 2.0%
Medals per Games: 1992-2004 / 2006-2022	54.4 / 66.8
Host medals at home / host average	44 / 50.4 over 16 Games
Women / men medals	492 / 546
Top-3 sport share	25.4%

How to refresh, change and rebuild the report

Refresh the data

Put `Project_06_Olympic_Games.xlsx` back in `C:\Users\USER\Downloads\` (or change the path in Power Query), open `Olympics.pbip` and press **Refresh**. Then update the numbers in titles and notes, which are written text.

Change a page

Pages are generated. Edit `_build\generate-report.js`, then run:

```
cd "C:\Users\USER\Desktop\ALL PROJECTS\OLympics"
node _build\generate-report.js
powerbi-report-author validate Olympics.Report
```

Close Power BI first (or apply external changes) so Desktop does not save its older copy over the new pages. Shared building blocks live in `ALL PROJECTS\_shared\pbir-kit.js`.

Change a measure

Edit it in Power BI Desktop and save, or edit `Olympics.SemanticModel\definition\tables\MedalTable.tmdl` with Desktop closed. `_build\fix-model.js` re-applies this version's fixes and is safe to run twice.

Change the cover

`_build\make-cover.ps1` draws the rings image (size, position and opacity are at the top of the script); the generator copies it into the report.

What the numbers can and cannot tell you

- **Treat this as a teaching dataset.** Real Games award a gold, silver and bronze in every event and sports belong to one season; here only 13% of event-Games have medals and every sport appears in both seasons.
- Comparisons between nations, regions and income groups still hold, because every nation is measured the same way.
- Season comparisons are not meaningful; age and body measures are not used.
- Population and GDP are single values at the latest Games, applied to all years.
- Greece (2004 host) is not in the nation list, so 16 Games enter the host comparison.
- The source workbook is missing; the data cannot be refreshed until it is restored.

APPENDIX A

Project files

Path (in ALL PROJECTS\OLympics)	What it is
`Olympics.pbip`	Open this in Power BI Desktop
`Olympics.SemanticModel\`	The data model as TMDL text files
`Olympics.Report\`	The report as PBIR files. Generated; do not edit by hand
`_brief\report-spec.md`	Approved report spec
`_build\generate-report.js`	Generator for the theme and all report pages
`_build\fix-model.js`	Re-applies the measure fixes and additions
`_build\make-cover.ps1`	Draws the faded Olympic rings for the cover
`_build\assets\`	Author photo and cover image
`_build\backup\`	The original report and model before the redesign
`Olympics_Report_Guide.pdf`	This guide